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Patent Public Search | Text View

United States Patent Application Publication

20250278932

Kind Code

A1

Publication Date

September 04, 2025

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IMAGE DISPLAY SYSTEM, TERMINAL, METHOD, AND PROGRAM FOR DETERMINING A DIFFERENCE BETWEEN A FIRST IMAGE AND A SECOND IMAGE

Abstract

The present invention provides an image display system, a terminal, a method, and a program that can quickly and accurately display an image corresponding to a particular location. An image display system according to one example embodiment of the present invention includes: a current image acquisition unit configured to acquire a first image captured by a mobile terminal; an information acquisition unit configured to acquire information including a position and an orientation of the mobile terminal that has captured the first image; a past image acquisition unit configured to acquire a second image based on the position and the orientation of the mobile terminal and a position and an orientation associated with the second image stored in the past in a storage device; and a difference detection unit configured to detect a difference between the first image and the second image.

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Appl. No.: 19/212964

Filed: May 20, 2025

Foreign Application Priority Data

JP 2016-059612

Mar. 24, 2016

Related U.S. Application Data

parent US continuation 18237707 20230824 PENDING child US 19212964
parent US continuation 18140463 20230427 parent-grant-document US 12333797 child US 18237707
parent US continuation 17329787 20210525 parent-grant-document US 12039773 child US 18140463
parent US continuation 16086942 20180920 ABANDONED WO continuation PCT/JP2017/011654 20170323 child US 17329787

Publication Classification

Int. Cl.: **G06V20/00** (20220101); **G06F1/16** (20060101); **G06F16/587** (20190101); **G06V20/20** (20220101); **G06V20/52** (20220101); **H04N23/80** (20230101)

U.S. Cl.:

CPC **G06V20/00** (20220101); **G06F1/1686** (20130101); **G06F16/587** (20190101); **G06V20/20** (20220101); **G06V20/52** (20220101); **H04N23/80** (20230101);

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 18/237,707 filed Aug. 24, 2023, which is a continuation of U.S. application Ser. No. 18/140,463 filed Apr. 27, 2023, which is a continuation of U.S. application Ser. No. 17/329,787 filed May 25, 2021, which issued as U.S. Pat. No. 12,039,773, which is a continuation of U.S. application Ser. No. 16/086,942 filed Sep. 20, 2018, which is a National Stage of International Application No. PCT/JP2017/011654 filed Mar. 23, 2017, claiming priority based on Japanese Patent Application No. 2016-059612, filed Mar. 24, 2016, the disclosure of which is incorporated herein in its entirety by reference.

TECHNICAL FIELD

[0002] The present invention relates to an image display system, a terminal, a method, and a program that display an image captured in the past at a particular location.

BACKGROUND ART

[0003] In security of an area, detection of a suspicious object is performed. A suspicious object may be an explosive or a toxic substance, for example, which is basically an object which is not usually placed at the place. Thus, detection of a suspicious object requires determination as to whether or not the object is usually placed.

[0004] In particular, in security of an important event, surveys are performed over multiple times, and the personnel may be different for each survey. For example, in security of an important person, personnel who perform a survey in advance on a planned route and personnel who perform safety check immediately before the important person passes by are often different in actual practice. It is difficult to report all, to the personnel who perform check immediately before an important person passes by, whether or not it is an object that has been confirmed to be usually placed. Thus, when a suspicious object has been placed after the previous survey, the personnel who check immediately before the important person passes by may not notice the suspicious object or may overreact to the object and take time for check even though it is not a suspicious object.

[0005] For example, it may be possible to capture and accumulate various places as images in advance in a prior survey, compare the images with the current landscape in the prior survey, and thereby examine whether or not an object which was not present in the prior survey increases in the

current landscape. In this case, such a technology is required that stores and provides many images captured in the prior survey in association with capturing places.

[0006] In the art disclosed in Patent Literature 1, images associated with position information indicating capturing places are accumulated on a server, and the server selects an image close to the current position or a position designated by a user out of the accumulated images to display the image on a display of a mobile terminal.

CITATION LIST

Patent Literature

[0007] PTL 1: Japanese Patent Application Laid-Open No. 2009-187233

Non Patent Literature

[0008] NPL 1: “Quasi-Zenith Satellites System User Interface Specification,” [online], Nov. 28, 2014, Japan Aerospace Exploration Agency, [searched on Feb. 25, 2016], Internet <URL: http://qz-vision.jaxa.jp/USE/is-qzss/DOCS/IS-QZSS_16_J.pdf>, p. A1-A12

[0009] NPL 2: “Image Recognition Service “GAZIRU” [online], NEC Corporation, [searched on Feb. 29, 2016], Internet <URL: <http://jpn.nec.com/solution/cloud/gazou/pdf/GAZIRU.pdf>>

SUMMARY OF INVENTION

[0010] Since the art disclosed in Patent Literature 1 only displays an image captured at a position close to the current position or a designated position, however, it is not to display an image in which a particular place (landscape) is captured. That is, an image acquired based only on a position may include a plurality of images in which all the directions are captured from the position. It takes much time to search many images acquired based only on a position for an image in which a particular place is captured. In security performed in a wide range, since it is demanded to reduce time taken at one place, it is difficult to use the art disclosed in Patent Literature 1 for security including detection of a suspicious object.

[0011] The present invention has been made in view of the above problem and intends to provide an image display system, a terminal, a method, and a program that can quickly and accurately display an image corresponding to a particular place.

[0012] The first example aspect of the present invention is an image display system including: a first image acquisition unit configured to acquire a first image captured by a mobile terminal; an information acquisition unit configured to acquire information including a position and an orientation of the mobile terminal that has captured the first image; a second image acquisition unit configured to, based on the position and the orientation of the mobile terminal and a position and an orientation associated with a second image stored in the past in a storage device, acquire the second image; and a difference detection unit configured to detect a difference between the first image and the second image.

[0013] The second example aspect of the present invention is a terminal including: a first image acquisition unit configured to acquire a first image captured; an information acquisition unit configured to acquire information including a position and an orientation; and a difference detection unit configured to receive a second image acquired based on the position and the orientation and a position and an orientation associated with the second image stored in the past in a storage device and detect a difference between the first image and the second image.

[0014] The third example aspect of the present invention is a method including: acquiring a first image captured by a mobile terminal; acquiring information including a position and an orientation of the mobile terminal that has captured the first image; based on the position and the orientation of the mobile terminal and a position and an orientation associated with a second image stored in the past in a storage device, acquiring the second image; and detecting a difference between the first image and the second image.

[0015] The fourth example aspect of the present invention is a program configured to cause a computer to execute: acquiring a first image captured by a mobile terminal; acquiring information including a position and an orientation of the mobile terminal that has captured the first image;

based on the position and the orientation of the mobile terminal and a position and an orientation associated with a second image stored in the past in a storage device, acquiring the second image; and detecting a difference between the first image and the second image.

Advantageous Effects of Invention

[0016] According to the present invention, since an image stored in the past is acquired based on the position and the orientation of a mobile terminal that has captured a first image, it is possible to easily acquire and display a second image corresponding to a particular place which the user having the mobile terminal faces.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0017] FIG. 1 is a schematic diagram of an image display system according to a first example embodiment.

[0018] FIG. 2 is a general configuration diagram of a mobile terminal and a server according to the first example embodiment.

[0019] FIG. 3 is a schematic diagram illustrating a position of a user having the mobile terminal according to the first example embodiment.

[0020] FIG. 4 is a front view of a display of the mobile terminal displaying a current image.

[0021] FIG. 5 is a front view of a display of the mobile terminal displaying a past image.

[0022] FIG. 6 is a block diagram of an image display system according to the first example embodiment.

[0023] FIG. 7 is a diagram illustrating a flowchart of an image display method according to the first example embodiment.

[0024] FIG. 8 is a front view of a display of a mobile terminal displaying peripheral information.

[0025] FIG. 9 is a block diagram of an image display system according to a second example embodiment.

[0026] FIG. 10 is a diagram illustrating a flowchart of an image display method according to the second example embodiment.

[0027] FIG. 11 is a front view of a display of a mobile terminal accepting region designation.

[0028] FIG. 12 is a front view of a display of a mobile terminal displaying an object recognition result.

[0029] FIG. 13 is a block diagram of an image display system according to a third example embodiment.

[0030] FIG. 14 is a diagram illustrating a flowchart of an image display method according to the third example embodiment.

[0031] FIG. 15 is a front view of a display of a mobile terminal displaying a past image acquired in accordance with a designation condition.

[0032] FIG. 16 is a block diagram of an image display system according to a fourth example embodiment.

[0033] FIG. 17 is a diagram illustrating a flowchart of an image display method according to the fourth example embodiment.

[0034] FIG. 18 is a general configuration diagram of an image display system according to each example embodiment.

DESCRIPTION OF EMBODIMENTS

[0035] While example embodiments of the present invention will be described below with reference to the drawings, the present invention is not limited to the present example embodiments. Note that, in the drawings described below, components having the same function are labeled with the same references, and the duplicated description thereof may be omitted.

First Example Embodiment

[0036] FIG. 1 is a schematic diagram of an image display system **100** according to the present example embodiment. The image display system **100** has a mobile terminal **110** and a server **150**. The mobile terminal **110** and the server **150** are connected to each other via wired connection or wireless connection via a network **190** such as the Internet. The image display system **100** may include devices such as another server, another mobile terminal, or the like. The mobile terminal **110** is a terminal that is carried by a customer and has an image capture function using a camera and a function of displaying an image acquired from the server **150**. The server **150** is a server that is installed at any location and performs storage and acquisition of an image.

[0037] FIG. 2 is a general configuration diagram of the mobile terminal **110** and the server **150** according to the present example embodiment. Note that each of the mobile terminal **110** and the server **150** may be formed of a single device or may be formed of two or more physically separated devices in wired connection or wireless connection. For example, the server **150** is a laptop computer, a desktop computer, a workstation, a personal digital assistant, a server, a blade server, a mainframe, an embedded system, or the like. For example, the mobile terminal **110** is a personal digital assistant, a mobile phone, a smartphone, or the like. The specific hardware configuration of the mobile terminal **110** and the server **150** is not limited to the following configuration, and may be of various types or forms.

[0038] The server **150** has a processor **151**, memory **152**, and a storage device **153**. Further, the server **150** has a high-speed controller **154** including a high-speed interface and a low-speed controller **156** including a low-speed interface. The memory **152** and a high-speed expansion port **155** are connected to the high-speed controller **154**. Further, a display device such as a display **158** or the like is connected to the high-speed controller **154**. On the other hand, a low-speed expansion port **157** and a storage device **153** are connected to the low-speed controller **156**.

[0039] The processor **151**, the memory **152**, the storage device **153**, and the high-speed controller **154**, the low-speed controller **156**, and the high-speed expansion port **155** are connected to each other through various buses. Further, the processor **151**, the memory **152**, the storage device **153**, the high-speed controller **154**, the low-speed controller **156**, the high-speed expansion port **155**, and the low-speed expansion port **157** may be implemented on a common motherboard or may be implemented in other forms as appropriate.

[0040] The processor **151** is a central processing unit (CPU), for example, and is able to process instructions executed within the server **150**. Such instructions include an instruction that is used for displaying graphics information of a graphical user interface (GUI) on a display device such as the display **158** and stored in the memory **152** or the storage device **153**.

[0041] Further, a plurality of processors, a plurality of busses, or a plurality of processors and a plurality of busses can be used as appropriate together with a plurality of memory devices and multiple types of memory devices. Further, a plurality of servers **150** can be connected to each device that performs a part of the necessary process. For example, a plurality of servers **150** can be connected to each other as a server bank, a group of blade servers, or a multiprocessor system.

[0042] The memory **152** stores therein information within the server **150**. For example, the memory **152** may be a volatile memory unit or a non-volatile memory unit. The memory **152** may be another computer readable storage medium, such as a magnetic disk, an optical disk, or the like, for example.

[0043] The storage device **153** can configure mass storage used for the server **150**. The storage device **153** may be, for example, a computer readable storage medium such as a floppy (registered trademark) disk device, a hard disk device, an optical disk device, a tape device, a solid state memory device such as a flash memory, a disk array, or the like or may include such a computer readable storage medium. The storage device **153** may be a storage area network or a device with another configuration.

[0044] The high-speed controller **154** manages processes in which the bandwidth for the server **150**

is intensively used. On the other hand, the low-speed controller **156** manages processes in which the bandwidth is less intensively used. However, such allocation of the functions is a mere example, and allocation is not limited thereto. Further, a part or a whole of the high-speed controller **154** may be incorporated in the processor **151**.

[0045] The high-speed controller **154** is connected to the high-speed expansion port **155** that can accept the memory **152** and various expansion cards. Further, the high-speed controller **154** is connected to the display **158** via a graphics processor or an accelerator, for example.

[0046] The low-speed controller **156** is connected to the storage device **153** and the low-speed expansion port **157**. The low-speed expansion port **157** can include, for example, a communication port of various standards such as Universal Serial Bus (USB), Bluetooth (registered trademark), wired or wireless Ethernet (registered trademark), or the like. One or plurality of input/output devices such as a keyboard, a pointing device, a scanner, or the like can be connected to the low-speed expansion port **157**. Further, one or plurality of network devices such as a switch, a router, or the like can be connected to the low-speed expansion port **157** via a network adapter, for example. That is, the low-speed expansion port **157** functions as a communication interface.

[0047] The server **150** can be implemented in many forms without being limited to the forms described above. For example, the server **150** can be implemented in a form of a typical server or a plurality of servers in a form of a group of such servers. Further, the server **150** can be implemented as a part of the rack server system. Furthermore, the server **150** can be implemented in a form of a personal computer such as a laptop computer, a desktop computer, or the like.

[0048] Note that a part or a whole of the program executed by the processor **151** of the server **150** can be provided by a computer readable storage medium storing the above, such as a digital versatile disc-read only memory (DVD-ROM), a compact disc-read only memory (CD-ROM), a USB memory, or a flash memory.

[0049] The mobile terminal **110** has a processor **111**, memory **112**, a display **113**, a communication interface **114**, a camera **115**, and an electronic compass **116**. The mobile terminal **110** may have a storage device such as a micro drive, a flash memory, or the like as additional storage. Respective components of the mobile terminal **110** are connected to each other using various busses. At least some of the components of the mobile terminal **110** may be attached on a common motherboard or may be attached by other ways.

[0050] The processor **111** can execute instructions residing in the mobile terminal **110** including instructions stored in the memory **112**. The processor **111** may be implemented as a chip or a chipset having one or a plurality of analog or digital processors. The processor **111** controls other components of the mobile terminal **110** and performs display of a user interface, execution of an application, wireless communication, or the like, for example.

[0051] The processor **111** can perform input/output of information to a user through a display interface **119** and an input interface **120** connected to the display **113**. As a display **113**, any display unit such as a liquid crystal display, an organic electro luminescence (EL) display, or the like may be used, for example. The display interface **119** has a circuit necessary for driving the display **113** so as to present an image and other information to the user. Further, the display **113** is configured as a touch panel and outputs a user operation as an electric signal. The input interface **120** accepts an electric signal indicating a user operation on the display **113** and converts it for transmission to the processor **111**.

[0052] The memory **112** stores therein information within the mobile terminal **110**. The memory **112** is a computer readable storage medium, a volatile memory unit, a non-volatile memory unit, or a combination thereof, for example. As the memory **112**, a flash memory, a non-volatile random access memory (NVRAM), or the like may be used, for example.

[0053] The camera **115** is an image capturing unit that outputs a resulted captured image as digital data. As the camera **115**, any image capturing device such as a charge coupled device (CCD) camera, a complementary metal oxide semiconductor (CMOS) camera, or the like may be used.

The camera **115** includes an image capturing element, an electric circuit, a lens, and the like necessary for image capturing.

[0054] The electronic compass **116** is an orientation acquisition unit that acquires the orientation of the mobile terminal **110**. As the electronic compass **116**, any device such as a terrestrial magnetism that is capable of acquiring the orientation of the mobile terminal **110** may be used.

[0055] The communication interface **114** is connected to a Global Positioning System (GPS) receiver **117** and a mobile communication device **118**. The mobile communication device **118** performs mobile communication using any mode or protocol including a voice communication with Global System for Mobile communication (GSM) (registered trademark) scheme or the like, messaging such as Short Message Service (SMS), and data communication with Code Division Multiple Access (CDMA) scheme or the like. The mobile communication device **118** includes a processor, an electric circuit, an antenna, and the like necessary for mobile communication.

[0056] The GPS receiver **117** is a position acquisition unit that acquires the position of the mobile terminal **110**. The GPS receiver **117** includes a processor, an electric circuit, an antenna, and the like necessary for receiving the position of the mobile terminal **110** from the GPS. While the coordinates (x-coordinate and y-coordinate with a predetermined location being the origin) in the horizontal direction in a plan map display are used as a position for simplicity in the present example embodiment, any positional expression such as a latitude and a longitude may be used instead thereof. Furthermore, a height may be used as a position in addition to the coordinates in the horizontal direction. A height (altitude) can be calculated based on a GPS signal. This allows for distinction of a position in the height direction (for example, a level within a building), which makes it possible to acquire an image that is different for respective heights in addition to the horizontal direction coordinates in the image display method described below.

[0057] Since a position is calculated based on a signal from a satellite in the GPS, the position may not be acquired inside a building. Thus, a positioning technology utilizing Indoor Messaging System (IMES) (Non Patent Literature 1), a wireless Local Area Network (LAN), or the like that acquires a position by receiving a signal of position information transmitted from a transmitter provided on the land may be used instead of or in addition to the GPS.

[0058] The mobile terminal **110** can perform communication by voice using an audio interface **121**. The audio interface **121** is connected to a microphone and a speaker. The audio interface **121** accepts speech information from the user through the microphone and converts it into digital information which can be processed by the processor **111**. Further, the audio interface **121** can generate an audible sound to the user through the speaker, for example. A sound output by the audio interface **121** includes a voice from a voice call, a sound recorded in the memory **112** (for example, a voice message, a music file, or the like), or a sound generated by an application running on the mobile terminal **110**.

[0059] The mobile terminal **110** is provided with an expansion interface **122** and an external interface **123**. The expansion interface **122** is connected to expansion memory, for example. The expansion memory functions as additional storage to the mobile terminal **110** and may store a program running on the mobile terminal **110** or other information. Further, the expansion memory may store secure information and, in this case, functions as a security module.

[0060] The external interface **123** is configured to communicate with the processor **111** and can perform wired communication or wireless communication. For example, the external interface **123** performs near field communication such as Bluetooth (registered trademark) or wireless communication such as wireless LAN between the mobile terminal **110** and other devices. As the external interface **123**, a plurality of interfaces may be combined and used.

[0061] FIG. 3 is a schematic diagram illustrating the position of a user having the mobile terminal **110**. The user directs the camera **115** of the mobile terminal **110** to a particular orientation A2 at a position A1. In this state, the mobile terminal **110** sequentially displays the image captured by the camera **115** (referred to as a current image or a first image) on the display **113**. Since the user

usually holds the camera **115** in front thereof to capture an image, an image captured range of the current image corresponds to a current field of view of the user having the mobile terminal **110**. In the following, the position and the orientation of the mobile terminal **110** in a state where the user holds the camera **115** in front thereof are referred to as a current position and a current orientation of the mobile terminal **110**. Further, in response to the user performing a predetermined operation with the mobile terminal **110**, the mobile terminal **110** displays an image including a place captured in the past which is similar to the current image (referred to as a past image or a second image) on the display **113**.

[0062] FIG. **4** is a front view of the display **113** of the mobile terminal **110** displaying a current image **B1**. Upon startup of an image display program according to the present example embodiment, the mobile terminal **110** enters a state of displaying the current image on the display **113** as illustrated in FIG. **4**. The display **113** displays the current image **B1**, a text **B4** indicating that it is the current image, and a button **B5** used for displaying the past image.

[0063] The current image **B1** is an image captured by the camera **115**. For example, an image of an object **B2** is included within the current image **B1**. A frame **B3** indicating a difference from a past image in which a similar place is captured is further included in the current image **B1**. A method of detection of the difference between the current image and the past image will be described later. In the example of FIG. **4**, since the object **B2** is not included in the past image (FIG. **5**) and thus detected as a difference, the frame **B3** surrounding a region including the object **B2** is displayed.

[0064] The button **B5** is a virtual button displayed on the display **113** and can be pressed by the user operating a touch panel or the like. In response to the button **B5** being pressed by the user, the mobile terminal **110** transfers to a state where the past image is displayed on the display **113** as illustrated in FIG. **5**.

[0065] FIG. **5** is a front view of the display **113** of the mobile terminal **110** displaying the past image **B6**. The display **113** displays a past image **B6**, a text **B8** indicating capturing time of the past image **B6**, and a button **B9** used for displaying the current image.

[0066] The past image **B6** is a past image received from the server **150** and associated with the position and orientation similar to the current position and orientation. A selection method of the past image **B6** will be described later. The text **B8** indicating capturing time of the past image **B6** is received from the server **150** together with the past image **B6**.

[0067] A frame **B7** indicating a difference from the current image is included within the past image **B6**. The frame **B7** of the past image **B6** is displayed at a position corresponding to the frame **B3** in the current image **B1**. In comparison between the current image **B1** and the past image **B6** in which the difference thereof is emphasized in such a way, the user can determine that the object **B2** in the current image **B1** is newly arranged and is likely to be a suspicious object.

[0068] The button **B9** is a virtual button displayed on the display **113** and can be pressed by the user operating the touch panel or the like. In response to the button **B9** being pressed by the user, the mobile terminal **110** transfers to a state of displaying the current image on the display **113** as illustrated in FIG. **4**.

[0069] FIG. **6** is a block diagram of the image display system **100** according to the present example embodiment. In FIG. **6**, arrows represent main dataflows, and there may be other dataflows than those illustrated in FIG. **6**. In FIG. **6**, each block illustrates a configuration in a unit of function rather than in a unit of hardware (device).

[0070] The mobile terminal **110** has a current image acquisition unit **1101**, an information acquisition unit **1102**, a difference detection unit **1103**, and an image display unit **1104**. In the mobile terminal **110**, the current image acquisition unit **1101**, the information acquisition unit **1102**, the difference detection unit **1103**, and the image display unit **1104** are stored in the memory **112** as programs that can be executed by the processor **111**, respectively. That is, in execution of the image display method according to the present example embodiment, the processor **111** functions as the current image acquisition unit **1101**, the information acquisition unit **1102**, the difference detection

unit **1103**, and the image display unit **1104**. At least a part of these functions may be implemented as an electric circuit instead of a program.

[0071] The server **150** has a similarity degree calculation unit **1501**, a past image acquisition unit **1502**, and an image storage unit **1503**. In the server **150**, the similarity degree calculation unit **1501** and the past image acquisition unit **1502** are stored in the memory **152** or the storage device **153** as programs that can be executed by the processor **151**. Further, the image storage unit **1503** is a part of the memory **152** or the storage device **153**. That is, in execution of the image display method according to the present example embodiment, the processor **151** functions as the similarity degree calculation unit **1501** and the past image acquisition unit **1502**, and the memory **152** or the storage device **153** functions as the image storage unit **1503**. At least a part of these functions may be implemented as an electric circuit instead of a program.

[0072] In response to a program of the image display method according to the present example embodiment being executed by the user, the current image acquisition unit **1101** (first image acquisition unit) captures a current image by using the camera **115** and stores the captured image in the memory **112**.

[0073] The information acquisition unit **1102** acquires the current position of the mobile terminal **110**, that is, the position at the time of acquisition of the current image by using the GPS receiver **117**. In the present example embodiment, for simplicity, the coordinates in the horizontal direction in a plan map display (x-coordinate and y-coordinate about a predetermined location as the origin) are used as a position. Alternatively, any positional expression such as a latitude and a longitude may be used as a position. Furthermore, a height may be used as a position in addition to the coordinates in the horizontal direction. At the same time, the information acquisition unit **1102** acquires the current orientation of the mobile terminal **110**, that is, the orientation at the time of acquisition of the current image by using the electronic compass **116**. An orientation is a direction in which the camera **115** captures an image. In the present example embodiment, for simplicity, an angle in the direction horizontal to the ground (azimuth angle) is used as an orientation. In addition to an azimuth angle, an angle of the direction perpendicular to the ground (elevation/depression angle) may be used as an orientation. Then, the information acquisition unit **1102** uses the mobile communication device **118** to transmit information including the acquired current position and orientation to the server **150**. The information acquisition unit **1102** acquires the current position and orientation every predetermined time (for example, every 1 second) and transmits information including the acquired current position and orientation to the server **150** successively or at the timing of a change in the current position and orientation.

[0074] The image storage unit **1503** pre-stores past images and the positions and the orientations of the captured ranges of the past images. In the following, a position and an orientation of a captured range of a past image are simply referred to as a position and an orientation of a past image. The image storage unit **1503** stores past images as image files that are electronic data. Each past image is provided with a unique identifier (for example, a file name). The image storage unit **1503** further stores the positions and the orientations of the past images and the capturing time of the past images as a table of a database, for example, in association with the identifiers of the past images.

[0075] The similarity degree calculation unit **1501** uses the current position and orientation received from the mobile terminal **110** and the position and orientation associated with past images pre-stored in the image storage unit **1503** to calculate a similarity degree indicating the resemblance between them based on a predetermined calculation criterion.

[0076] A calculation method of a similarity degree performed by the similarity degree calculation unit **1501** will be described below. The similarity degree calculation unit **1501** calculates a similarity degree S by using Equation (1) for respective past images pre-stored in the image storage unit **1503**.

$$S = (P_0 - P)^2 + (Q_0 - Q)^2 \quad (1)$$

[0077] The value P_0 denotes the current orientation, and the value P denotes the orientation of a past image. The value Q_0 denotes the current position, and the value Q denotes the position of the past image. The value α is a weighting factor used for adjusting weighting of the orientation and the position. Since the orientation is expressed by an azimuth angle in the present example embodiment, the values P_0 and P each are an angle. Since the position is expressed by x-y coordinates in the present example embodiment, the values Q_0 and Q are two-dimensional coordinates, respectively, and the item $(Q_0 - Q)^2$ means the square of the distance between the current position and the position of the past image.

[0078] According to Equation (1), the closer the position and orientation of a target past image to the current position and orientation of the mobile terminal **110** is, the lower the similarity degree S is. That is, the similarity degree S is an index indicating the resemblance of the position and orientation between the target past image and the mobile terminal. The definition of the similarity degree S described here is an example, and any definition that represents the resemblance between the current position and orientation and the position and orientation of a past image may be used. While a lower similarity degree S corresponds to higher resemblance in Equation (1), the similarity degree S may be defined in the opposite manner.

[0079] The similarity degree calculation unit **1501** may calculate the similarity degree S for all the past images stored in the image storage unit **1503** or may calculate the similarity degree S for some past images which satisfy a condition. For example, the amount of calculation can be reduced with only the past images having a position which has a distance to the current position of the mobile terminal **110** below a predetermined value being a calculation target.

[0080] The past image acquisition unit **1502** (second image acquisition unit) acquires, from the image storage unit **1503**, a past image having the smallest similarity degree S (that is, having the highest resemblance) out of the past images whose similarity degrees S have been calculated. The past image acquisition unit **1502** then uses the communication interface to transmit the acquired past image to the mobile terminal **110**.

[0081] The difference detection unit **1103** detects a difference between a current image captured by the camera **115** and a past image received from the server **150**. As a detection method of a difference, a known method may be used. For example, the difference detection unit **1103** first extracts feature points of the current image and the past image, respectively, and performs positioning of the current image and the past image to each other so that the feature points match each other. As a feature point, a point (edge) in a current image and a past image at which a brightness or a color shape changes may be used. In order to improve the accuracy of positioning, correction such as change of a color tone, an angle, a position, and a magnification may be performed in advance on the current image and the past image.

[0082] Next, the difference detection unit **1103** calculates a difference in brightness between pixels at the same position in the current image and the past image on which positioning has been performed. Out of regions having continuous pixels which have a difference in brightness value above a predetermined value, the difference detection unit **1103** then detects those having the area above a predetermined value as a difference of the current image and the past image. The difference value in brightness value used in detection of a difference and the area value may be determined by an experiment or a simulation as appropriate. The difference detection unit **1103** may detect one or a plurality of differences from a current image and a past image.

[0083] Next, a current image and a past image input to the difference detection unit **1103** are acquired based on a position and an orientation and thus are well resemble to each other. Therefore, since the difference detection unit **1103** can perform position matching and detection of a difference starting from a state where the positions and the orientations of the current image and the

past image are close to each other, the difference can be accurately detected.

[0084] The difference detection unit **1103** determines each size and each position of the frame B3 displayed on the current image and the frame B7 displayed in the past image based on the size and the position of the detected difference. When the difference detection unit **1103** has performed correction such as change of an angle, a position, and a magnification on the current image and the past image in positioning, it compensates the correction and determines the size and the position of the frames B3 and B7. For example, when the current image and the past image are relatively moved because of positing, each size and each position of the frames B3 and B7 are calculated so as to indicate the difference in the current image and the past image on the current image and the past image that have not been moved.

[0085] While provided to the mobile terminal **110** in the preset example embodiment, the difference detection unit **1103** may be provided to the server **150**. In this case, the server **150** transmits a difference detection result from the difference detection unit **1103** to the mobile terminal **110**.

[0086] The image display unit **1104** can be in a current image display state as illustrated in FIG. 4 and a past image display state as illustrated in FIG. 5 and displays the current image or the past image on the display **113**. The current image display state and the past image display state are switched by the buttons B5 and B9 on the display **113** being pressed.

[0087] In the current image display state, the image display unit **1104** displays the current image B1 acquired by the current image acquisition unit **1101** on the display **113** as illustrated in FIG. 4. At the same time, the image display unit **1104** displays the frame B3 indicating the difference determined by the difference detection unit **1103** overlapped with the current image B1. When no difference is detected by the difference detection unit **1103**, the image display unit **1104** does not display the frame B3.

[0088] In the past image display state, the image display unit **1104** displays the past image B6 received from the server **150** on the display **113** as illustrated in FIG. 5. At the same time, the image display unit **1104** displays the frame B7 indicating the difference determined by the difference detection unit **1103** overlapped with the past image B6. When no difference is detected by the difference detection unit **1103**, the image display unit **1104** does not display the frame B7.

[0089] A display method of the difference described above is an example, and any method that enables the user to recognize a difference may be used to display the difference. While the image display unit **1104** displays a difference in both of the current image B1 and the past image B6 in the present example embodiment, a difference may be displayed in at least one of the current image B1 and the past image B6.

[0090] The frame B3 and the frame B7 indicating a difference are not limited to the rectangular as illustrated in FIG. 4 and FIG. 5, but may be any shape such as a polygon, a circle, or the like. Instead of or in addition to the frame B3 and the frame B7 indicating a difference, the image display unit **1104** may display a difference by colors in the current image B1 or the past image B6. In this case, it is possible to display the difference by changing the color between a region including a difference and other regions, for example, by representing a region including a difference in the current image B1 or the past image B6 with color display and another region with monochrome display.

[0091] Instead of representing a frame indicating a difference in the current image B1 or the past image B6, the image display unit **1104** may display a difference by displaying only a partial region including the difference in the current image B1 or the past image B6 on the display **113**.

[0092] Not only a past image stored on a position and orientation basis but also a panoramic image captured for a wider angle range may be used as a past image stored in the image storage unit **1503**. A panoramic image includes an image captured with an angle range wider than a captured range of the common camera **115** at a single position or an image captured with the entire range of 360 degrees. The panoramic image is created by combining a plurality of images captured in different

orientations at the same position, for example. When a panoramic image close to the current position of the mobile terminal **110** is present, the similarity degree calculation unit **1501** may select, as a past image, a part cut out from the panoramic image in a direction corresponding to the current orientation of the mobile terminal **110**. Further, the similarity degree calculation unit **1501** may select the whole panoramic image as a past image. In this case, the past image display unit **1104** may display a portion in a direction corresponding to the current orientation of the mobile terminal **110** out of the received panoramic image.

[0093] By selecting a past image using not only a position but also an orientation in such a way, the mobile terminal **110** can easily display a past image corresponding to the current captured range of the camera **115** (that is, the user's field of view). Furthermore, since a current image and a past image similar to each other can be acquired based on the position and the orientation, the mobile terminal **110** can accurately detect a difference between a current image and a past image. The user can easily find a suspicious object based on the displayed difference.

[0094] FIG. 7 is a diagram illustrating a flowchart of an image display method according to the present example embodiment. The image display method is performed in association with the mobile terminal **110** and the server **150**. The image display method is started when a predetermined operation such as pressing of a button is detected on the mobile terminal **110**, for example.

[0095] First, the current image acquisition unit **1101** captures a current image using the camera **115** and stores the captured image in the memory **112** (step **S11**). Next, the information acquisition unit **1102** acquires the current position of the mobile terminal **110** by using the GPS receiver **117** and acquires the current orientation of the mobile terminal **110** by using the electronic compass **116** (step **S12**). The information acquisition unit **1102** uses the mobile communication device **118** to transmit information including the acquired current position and orientation to the server **150**. The order of step **S11** and step **S12** may be opposite or may be performed in parallel.

[0096] The similarity degree calculation unit **1501** uses the current position and orientation received from the mobile terminal **110** to calculate the similarity degree S to each past image pre-stored in the image storage unit **1503** based on Equation (1) described above, for example (step **S13**).

[0097] The past image acquisition unit **1502** acquires, from the image storage unit **1503**, one past image having the smallest similarity degree S out of the past images whose similarity degrees S have been calculated in step **S12** (step **S14**). The past image acquisition unit **1502** uses the communication interface to transmit the acquired past image to the mobile terminal **110**.

[0098] The difference detection unit **1103** detects a difference between the current image acquired in step **S11** and the past image acquired in step **S14** (step **S15**). A detection result of the difference from the difference detection unit **1103** includes a size and a position of a frame displayed on the current image or the past image so as to display the difference.

[0099] If a button for displaying the current image is pressed at the start of this flowchart (step **S16**, YES), the image display unit **1104** displays a frame indicating the difference detected in step **S15** on the display **113** in addition to the current image captured by the camera **115** (step **S17**).

[0100] If a button for displaying the current image is not pressed at the start of this flowchart, that is, a button for displaying a past image is pressed (step **S16**, NO), the image display unit **1104** displays a frame indicating the difference detected in step **S15** on the display **113** in addition to the past image received from the server **150** (step **S18**).

[0101] The processor **111** of the mobile terminal **110** and the processor **151** of the server **150** are subjects of each step (process) included in the image display method illustrated in FIG. 7. That is, the processor **111** and the processor **151** read the program for executing the image display method illustrated in FIG. 7 from the memory **112**, the memory **152**, or the storage device **153**, execute the program to control respective units of the mobile terminal **110** and the server **150**, and thereby perform the image display method illustrated in FIG. 7.

[0102] In the present example embodiment, a current image and a past image corresponding thereto

may be simply displayed without detection and display of a difference. In this case, the user visually examines a difference between the current image and the past image to find a suspicious object.

[0103] According to the image display system **100** of the present example embodiment, the mobile terminal **110** can easily display a past image captured at a place similar to a current captured range of the camera **115** by only directing the camera **115**. If an image is acquired by using only the position as seen in the technology disclosed in Patent Literature 1, since images captured in different directions from the same location cannot be distinguished, it is not possible to select an image corresponding to the current user's field of view. In contrast, the image display system **100** according to the present example embodiment acquires a past image by calculating a similarity degree using not only the position but also the orientation of the mobile terminal **100** and thus can display a past image corresponding to the current captured range of the camera **115**, that is, the user's field of view. Furthermore, since the image display system **100** can automatically display the difference between the current image and the past image, this enables the user to easily recognize a portion having a difference between the current image and the past image.

[0104] When the image display system **100** of the present example embodiment is used for security, it is possible to easily detect a suspicious object by quickly and accurately acquiring a past image corresponding to the current user's field of view and further identifying a difference between a current state and a past state.

[0105] The image display system **100** according to the present example embodiment can be applied to tourism application without being limited to security. It is possible to provide past images of different seasons or different time for the same landscape to the user by pre-storing images of different seasons, years, and months associated with the landscape as past images. Further, it is possible to provide to the user a picture in which the landscape is drawn by pre-storing a picture (landscape painting) associated with the landscape as a past image. In this case, the position and orientation of the drawing range of a picture can be used as the position and orientation associated with the past image. As discussed above, the image display system **100** can easily and accurately display an image associated with a particular place, that is, a particular position and orientation.

Second Example Embodiment

[0106] In the present example embodiment, a position and an orientation of a pre-stored past image is displayed on a map, and a user is then notified that there is a past image close to the current position. An image display system **200** according to the present example embodiment has the same device configuration as that of the first example embodiment.

[0107] FIG. **8** is a front view of the display **113** of the mobile terminal **110** displaying peripheral information. In response to startup of an image display program according to the present example embodiment, the mobile terminal **110** transfers to a state of displaying peripheral information on the display **113** as illustrated in FIG. **8**. The display **113** displays a map **C1**, a current marker **C2**, position markers **C3**, orientation markers **C4**, and a panorama marker **C5** as peripheral information.

[0108] The map **C1** is a map of a predetermined range about the current position of the mobile terminal **110** as the center. The current position of the mobile terminal **110** is acquired every predetermined time (for example, every 1 second) by using the GPS receiver **117**. The map **C1** may be a map stored in the memory **112** of the mobile terminal **110** or may be acquired at any time via the network **190**. The current marker **C2** is displayed at a point on the map **C1** corresponding to the current position of the mobile terminal **110**.

[0109] The position marker **C3** is displayed at a point on the map **C1** corresponding to the position of each past image. The past image is an image included in a predetermined range about the current position as the center out of past images pre-stored in the server **150**. Furthermore, at least one orientation marker **C4** indicating the orientation of the past image is displayed near respective position markers **C3** of past images. When a past image having a plurality of orientations at a single position is stored, a plurality of orientation markers **C4** may be displayed for a single position

marker **C3**. In the present example embodiment, the orientation marker **C4** has an arrow shape, and the pointing direction of the arrow corresponds to an orientation. The expression of an orientation by the orientation marker **C4** is not limited thereto but may be a figure, a character, or a combination thereof. Further, when a panoramic image for 360 degrees is stored as a past image, the panorama marker **C5** indicating the whole direction may be displayed instead of the orientation marker **C4**. While the panoramic marker **C5** has a circular arrow shape in the present example embodiment, it may be any expression without being limited thereto. The position and the orientation of the past image are received at the mobile terminal **110** from the server **150** every predetermined time or at a timing when the current position changes.

[0110] FIG. **9** is a block diagram of the image display system **200** according to the present example embodiment. In FIG. **9**, arrows represent main dataflows, and there may be other dataflows than those illustrated in FIG. **9**. In FIG. **9**, each block illustrates a configuration in a unit of function rather than in a unit of hardware (device).

[0111] The mobile terminal **110** has an information acquisition unit **2101**, a peripheral information display unit **2102**, and a proximity notification unit **2103**. In the mobile terminal **210**, the information acquisition unit **2101**, the peripheral information display unit **2102**, and the proximity notification unit **2103** are stored in the memory **112** as programs that can be executed by the processor **111**, respectively. That is, when performing the image display method according to the present example embodiment, the processor **111** functions as the information acquisition unit **2101**, the peripheral information display unit **2102**, and the proximity notification unit **2103**. At least a part of these functions may be implemented as an electric circuit instead of a program.

[0112] The server **150** has a peripheral information acquisition unit **2501** and an image storage unit **2502**. In the server **150**, the image acquisition unit **2502** is stored in the memory **152** or the storage device **153** as a program that can be executed by the processor **151**. Further, the image storage unit **2502** is a part of the memory **152** or the storage device **153**. That is, when performing the image display method according to the present example embodiment, the processor **151** functions as the peripheral information acquisition unit **2501**, and the memory **152** or the storage device **153** functions as the image storage unit **2502**. At least a part of these functions may be implemented as an electric circuit instead of a program.

[0113] In response to the program of the image display method according to the present example embodiment being performed by the user, the information acquisition unit **2101** acquires the current position of the mobile terminal **110** (if necessary, in addition, the current orientation). The definition and the acquisition method of the position and the orientation are the same as those in the first example embodiment. The information acquisition unit **2101** then uses the mobile communication device **118** to transmit information including the acquired current position to the server **150**. The information acquisition unit **2101** acquires the current position every predetermined time (for example, every 1 second) and transmits information including the acquired current position to the server **150** successively or at the timing of a change in the current position.

[0114] The image storage unit **2502** pre-stores past images and the positions and orientations of the past images. The image storage unit **2502** stores past images as image files that are electronic data. Each past image is provided with a unique identifier (for example, a file name). The image storage unit **2502** further stores the positions and the orientations of the past images and capturing time of past images as a table of a database, for example, in association with the identifiers of the past images.

[0115] The peripheral information acquisition unit **2501** extracts a past image(s) that is within a predetermined distance (for example, within 100 m) from the current position received from the mobile terminal **110** out of past images pre-stored in the image storage unit **2502** and acquires the identifier and the position and orientation of the extracted past image. The extraction criteria of the past image around the current position is not limited to the criterion of being within a predetermined distance, but any criteria that can indicate the past image being close to the current

position such as a criterion of a predetermined number in ascending order, a criterion of being within the same block as the current position (for example, within the block having an address of the same street number), or the like may be used. The peripheral information acquisition unit **2501** transmits peripheral information including the identifier and the position and orientation of the extracted past image to the mobile terminal **110**.

[0116] The peripheral information display unit **2102** uses the current position acquired by the information acquisition unit **2101** and the peripheral information received from the server **150** to display the peripheral information on the display **113**. Specifically, as illustrated in FIG. **8**, the peripheral information display unit **2102** displays the position marker **C3** and the orientation marker **C4** (or the panorama marker **C5**) indicating the position and orientation of the past image included in the peripheral information together with the current marker **C2** indicating the current position on the map **C1** read from the memory **112** or acquired via the network **190**.

[0117] A single position marker **C3** may be displayed for a single past image position, or a single position marker **C3** may be displayed for a plurality of past image positions close to each other (for example, within 5 m). When a single position marker **C3** is displayed for a plurality of past images, the center of gravity of the plurality of past image positions may be the position of the position marker **C3**. Further, a single orientation marker **C4** may be displayed for a single past image orientation, or a single orientation marker **C4** is displayed for a plurality of past image orientations close to each other (for example, within 90 degrees). When a plurality of past image orientations are gathered, for example, an arrow in the north direction may be used when the azimuth angle (the north direction is 0 degree) is greater than or equal to 0 degree and smaller than or equal to 45 degrees or the azimuth angle is greater than or equal to 315 degrees and smaller than 360 degrees, an arrow in the east direction may be used when the azimuth angle is greater than 45 degrees and smaller than 135 degrees, an arrow in the south direction may be used when the azimuth angle is greater than or equal to 135 degrees and smaller than 225 degrees, and an arrow in the west direction may be used when the azimuth angle is greater than or equal to 225 degrees and smaller than 315 degrees.

[0118] The proximity notification unit **2103** uses a peripheral information received from the server **150** to notify the user of proximity to a past image when there is a past image within a predetermined distance (for example, within 10 m) from the current position acquired from the information acquisition unit **2101**. For example, the proximity notification unit **2103** uses an audio interface **121** to output a sound. The sound may include a beep sound, a music, a voice, or the like. As a method of notifying proximity to a past image, without being limited to a sound output, any method such as display on the display **113**, light emission of a lamp, vibration of the main unit, or the like may be used.

[0119] At least one of display of peripheral information by the peripheral information display unit **2102** and notification of proximity by the proximity notification unit **2103** may be performed. Further, display of peripheral information by the peripheral information display unit **2102** and notification of proximity by the proximity notification unit **2103** may be performed in any order. For example, notification of proximity by the proximity notification unit **2103** may be performed during display of peripheral information by the peripheral information display unit **2102**.

Contrarily, display of peripheral information by the peripheral information display unit **2102** may be started after notification of proximity by the proximity notification unit **2103** is performed.

[0120] FIG. **10** is a diagram illustrating a flowchart of an image display method according to the present example embodiment. The image display method is performed in cooperation with the mobile terminal **110** and the server **150**. The image display method is started when a predetermined operation such as pressing of a button or the like on the mobile terminal **110** is detected, for example.

[0121] First, the information acquisition unit **2101** acquires a current position of the mobile terminal **110** by the GPS receiver **117** (step **S21**). The information acquisition unit **2101** uses the

mobile communication device **118** to transmit information including the acquired current position to the server **150**.

[0122] The peripheral information acquisition unit **2501** extracts a past image(s) that is within a predetermined distance (for example, within 100 m) from the current position received from the mobile terminal **110** out of past images pre-stored in the image storage unit **2502** and acquires the identifier and the position and orientation of the extracted past image (step S22). The peripheral information acquisition unit **2501** uses the communication interface to transmit peripheral information including the identifier and the position and orientation of the extracted past image to the mobile terminal **110**.

[0123] The peripheral information display unit **2102** uses the current position acquired by step S21 and the peripheral information received from the server **150** to display the peripheral information on the display **113** (step S23). The peripheral information is displayed as illustrated in FIG. 8, for example.

[0124] If any of the past image positions included in the peripheral information received from the server **150** is within a predetermined distance (for example, within 10 m) from the current position acquired in step S21 (step S24, YES), the proximity notification unit **2103** uses the audio interface **121** to output a sound and notifies the user of proximity to the past image position (step S25).

[0125] If all the past image positions included in the peripheral information received from the server **150** are distant more than a predetermined distance from the current position acquired in step S21 (step S24, NO) or if no finish instruction is received from the user after the completion of step S25 (step S26, NO), the process returns to step S21 and is repeated. On the other hand, if a finish instruction is received from the user (step S26, YES), the process according to the present example embodiment ends.

[0126] The processor **111** of the mobile terminal **110** and the processor **151** of the server **150** are subjects of each step (process) included in the image display method illustrated in FIG. 10. That is, the processor **111** and the processor **151** read the program for executing the image display method illustrated in FIG. 10 from the memory **112**, the memory **152**, or the storage device **153**, execute the program to control respective units of the mobile terminal **110** and the server **150**, and thereby perform the image display method illustrated in FIG. 10.

[0127] The image display method according to the present example embodiment can be used in combination with the first example embodiment. In this case, when the user has moved to the past image position confirmed on a map according to the present example embodiment or when the user is notified of there being a past image nearby, display of the difference between a current image and a past image on the mobile terminal may be started by the image display method according to the first example embodiment. Alternatively, the image display method according to the present example embodiment may be performed alone.

[0128] According to the present example embodiment, the user can know a past image position around the current location by viewing a map displayed on the display **113** or receiving notification from the mobile terminal **110**. This allows the user to positively move to a location where a past image is located and acquire a past image stored at the location or view the difference as described in the first example embodiment.

Third Example Embodiment

[0129] In the present example embodiment, information on an object included in a current image or a past image is displayed by using an object recognition technology. An image display system **300** according to the present example embodiment has the same device configuration as that of the first example embodiment.

[0130] FIG. 11 is a front view of the display **113** of the mobile terminal **110** accepting region designation. In response to startup of an image display program according to the present example embodiment, the mobile terminal **110** transfers to a state of accepting region designation on the display **113** as illustrated in FIG. 11. The display **113** displays a past image D1, a frame D3

indicating a designated region, and a button **D4** used for determining a region.

[0131] For example, the past image **D1** is acquired by the image display method according to the first example embodiment and received from the server **150**. The current image may be used instead of the past image **D1**. The frame **D3** indicating a region to be designated as object recognition is displayed in the past image **D1**. The position and the size of the frame **D3** can be changed in response to a user operation (for example, swipe on a touch panel). In the state of FIG. **11**, the frame **D3** is set so as to surround an object **D2** in the past image **D1**.

[0132] The button **D4** is a virtual button displayed on the display **113** and can be pressed by the user operating a touch panel or the like. In response to the button **D4** being pressed by the user, the mobile terminal **110** transfers to a state of displaying an object recognition result on the display **113** as illustrated in FIG. **12**.

[0133] FIG. **12** is a front view of the display **113** of the mobile terminal **110** displaying an object recognition result. The display **113** displays the past image **D1**, the frame **D3** indicating a designated region, an object recognition result **D5**, and a button **D6** used for returning to region determination. In a similar manner to FIG. **11**, the current image may be used instead of the past image **D1**.

[0134] The object recognition result **D5** determined by an object recognition method described later is displayed near the frame **D3** indicating the region determined in FIG. **11**. The object recognition result **D5** includes a text of the name of the object. The object recognition result **D5** indicating that the object **D2** is a public trash box is displayed here. The object recognition result **D5** may include any information on an object such as additional information on the object, an image of a similar object other than the name of the object, or the like.

[0135] The button **D6** is a virtual button displayed on the display **113** and can be pressed by the user operating a touch panel or the like. In response to the button **D6** being pressed by the user, the mobile terminal **110** again transfers to a state of accepting region designation on the display **113** as illustrated in FIG. **11**.

[0136] FIG. **13** is a block diagram of the image display system **300** according to the present example embodiment. In FIG. **13**, arrows represent main dataflows, and there may be other dataflows than those illustrated in FIG. **13**. In FIG. **13**, each block illustrates a configuration in a unit of function rather than in a unit of hardware (device).

[0137] The mobile terminal **110** has a region designation unit **3101** and an image display unit **3102**. In the mobile terminal **110**, the region designation unit **3101** and the image display unit **3102** are stored in the memory **112** as programs that can be executed by the processor **111**, respectively. That is, when performing the image display method according to the present example embodiment, the processor **111** functions as the region designation unit **3101** and the image display unit **3102**. At least a part of these functions may be implemented as an electric circuit instead of a program.

[0138] The server **150** has an object recognition unit **3501** and an object information storage unit **3502**. In the server **150**, the object recognition unit **3501** is stored in the memory **152** or the storage device **153** as programs that can be executed by the processor **151**. Further, the object information storage unit **3502** is a part of the memory **152** or the storage device **153**. That is, when performing the image display method according to the present example embodiment, the processor **151** functions as the object recognition unit **3501**, and the memory **152** or the storage device **153** functions as the object information storage unit **3502**. At least a part of these functions may be implemented as an electric circuit instead of a program.

[0139] In response to the program of the image display method according to the present example embodiment being performed by the user, the region designation unit **3101** accepts designation of a region as a target of object recognition on a current image or a past image. When a region is determined by the user, the region designation unit **3101** transmits an image region that is a specified portion cut out from the current image or the past image to the server **150**. Note that, when a storage unit for past images is provided in the server **150** that performs object recognition,

the mobile terminal **110** is not required to transmit an image region to the server **150**, and instead may transmit the position and the size of the designated region.

[0140] The object recognition unit **3501** uses a known object recognition method (for example, the method disclosed in Non Patent Literature 2) to perform recognition of an object included in the image region received from the mobile terminal **110**. For example, first, the object recognition unit **3501** extracts feature information (edge or the like) from the image region received from the mobile terminal **110**. Further, in addition to various object feature information, the name or additional information on the object is pre-stored in the object information storage unit **3502**. The object recognition unit **3501** then compares the feature information extracted from the image region with the feature information stored in the object information storage unit **3502** and determines the object included in the image region according to a predetermined rule. The object recognition method according to the present example embodiment is not limited to a specific method, and any method that can recognize an object from an image may be used. The object recognition unit **3501** transmits an object recognition result including the name or the like of the determined object to the mobile terminal **110**.

[0141] The object recognition unit **3501** may be provided in the mobile terminal **110** instead of in the server **150**. In such a case, it is not necessary to transmit an image region itself from the mobile terminal **110** to the server **150**, and it is necessary to transmit only the feature information extracted from the image region and the feature information on the object stored in the object information storage unit **3502**. As a result, the traffic necessary for the image display method according to the present example embodiment can be reduced.

[0142] The image display unit **3102** displays the past image **D1** (or the current image) and the frame **D3** indicating a designated region and displays the object recognition result **D5** received from the server **150** on the display **113** as illustrated in FIG. **12**.

[0143] FIG. **14** is a diagram illustrating a flowchart of the image display method according to the present example embodiment. The image display method is started when a predetermined operation such as pressing of a button or the like is detected on the mobile terminal **110**, for example.

[0144] First, in response to the program of the image display method according to the present example embodiment being performed by the user, the region designation unit **3101** accepts designation of a region as a target of object recognition on a current image or a past image (step **S31**). When a region is determined by the user, the region designation unit **3101** uses the mobile communication device **118** to transmit an image region that is a specified portion cut out from the current image or the past image to the server **150**.

[0145] The object recognition unit **3501** uses the object recognition method described above to perform recognition of an object included in the image region received from the mobile terminal **110** (step **S32**). The object recognition unit **3501** uses the communication interface to transmit an object recognition result including the name or the like of the determined object to the mobile terminal **110**.

[0146] The image display unit **3102** displays the object recognition result received from the server **150** on the display **113** (step **S33**). The object recognition result is displayed as illustrated in FIG. **12**, for example.

[0147] The processor **111** of the mobile terminal **110** and the processor **151** of the server **150** are subjects of each step (process) included in the image display method illustrated in FIG. **14**. That is, the processor **111** and the processor **151** read the program for executing the image display method illustrated in FIG. **14** from the memory **112**, the memory **152**, or the storage device **153**, execute the program to control respective units of the mobile terminal **110** and the server **150**, and thereby perform the image display method illustrated in FIG. **14**.

[0148] While an object recognition technology is applied to only the region designated by the user in a current image or a past image in the image display method according to the present example embodiment, the object recognition technology may be applied to the entire region in a current

image or a past image without acceptance of designation from the user, and an object recognition result may be automatically displayed.

[0149] The image display method according to the present example embodiment can be used in combination with the first example embodiment. In this case, designation of a region to be a target for object recognition by the present example embodiment may be accepted after a difference between a current image and a past image is displayed on the mobile terminal by using the image display method according to the first example embodiment. Alternatively, a region including a difference between a current image and a past image determined by the first example embodiment may be automatically defined as a region to be a target of the object recognition, and the object recognition according to the present example embodiment may be performed thereon.

Alternatively, the image display method according to the present example embodiment may be performed alone.

[0150] According to the present example embodiment, since information on an object included in a current image or a past image is acquired by using an object recognition technology, identification of a suspicious object can be easily performed. Further, when the present example embodiment is used in application of tourism, guide information such as description of an object included in a current image or a past image can be provided to the user.

Fourth Example Embodiment

[0151] In the present example embodiment, a past image is displayed according to a condition designated by the user. An image display system **400** according to the present example embodiment has the same device configuration as that in the first example embodiment.

[0152] FIG. **15** is a front view of the display **113** of the mobile terminal **110** displaying a past image acquired according to a designated condition. In response to startup of an image display program according to the present example embodiment, the mobile terminal **110** transfers to a state of displaying a condition designation part that accepts a condition designation of a past image to be displayed on the display **113** together with a past image as illustrated in FIG. **15**. The display **113** displays, together with a past image E1 and a text E2 indicating the capturing time of the past image E1, a time designation part E3, a tug designation part E4, and a free entry part E5 as condition designation parts.

[0153] The past image E1 is a past image received from the server **150** and associated with the position and orientation similar to the current position and orientation. A selection method of the past image E1 will be described later. The text E2 indicating the capturing time of the past image E1 is received from the server **150** together with the past image E1.

[0154] The time designation part E3 is a virtual slide bar for designation of time displayed on the display **113** and can be operated on the touch panel or the like by the user. The time designation part E3 according to the present example embodiment includes a knob movable in a lateral direction, and the position of the knob corresponds to the current setting value. The knob positioned closer to the left end indicates older time, and the knob positioned closer to the right end indicates later time. A specific form of the time designation part E3 is not limited thereto, and it may be a form in which time is designated by a numeric value.

[0155] The tug designation part E4 is a virtual button used for tug designation displayed on the display **113** and can be pressed by the user operating the touch panel or the like. Texts indicating one or more tugs associated with pre-stored past images are listed in the tug designation part E4 as buttons. A predetermined number of tugs of past images stored in the server **150** are included in the tug designation part E4. Displayed tugs may be selected by any criteria, for example, a predetermined number of tugs may be displayed in descending order of the number of past images stored in the server **150**, or a predetermined number of tugs associated with a past image close to the current location may be displayed.

[0156] The currently selected tug out of the tugs displayed in the tug designation part E4 is provided with a color different from that of other tugs (for example, “summer” is selected in FIG.

15). The free entry part E5 is a virtual entry field for tug designation displayed on the display 113 by which the user can enter a tug by operating a keyboard or the like. Thus, a tug which is not displayed on the tug designation part E4 can be designated from the free entry part E5.

[0157] When designated content of at least one of the time designation part E3, the tug designation part E4, and the free entry part E5 is changed by the user, the mobile terminal 110 acquires and displays a new past image E1 according to the designated content from the server 150.

[0158] FIG. 16 is a block diagram of the image display system 400 according to the present example embodiment. In FIG. 16, arrows represent main dataflows, and there may be other dataflows than those illustrated in FIG. 16. In FIG. 16, each block illustrates a configuration in a unit of function rather than in a unit of hardware (device).

[0159] The mobile terminal 110 has a condition acquisition unit 4101, an information acquisition unit 4102, and an image display unit 4103. In the mobile terminal 110, the condition acquisition unit 4101, the information acquisition unit 4102, and the image display unit 4103 are stored in the memory 112 as programs that can be executed by the processor 111, respectively. That is, when performing the image display method according to the present example embodiment, the processor 111 functions as the condition acquisition unit 4101, the information acquisition unit 4102, and the image display unit 4103. At least a part of these functions may be implemented as an electric circuit instead of a program.

[0160] The server 150 has a calculation target extraction unit 4501, a similarity degree calculation unit 4502, a past image acquisition unit 4503, an image storage unit 4504, and a tug storage unit 4505. In the server 150, the calculation target extraction unit 4501, the similarity degree calculation unit 4502, the past image acquisition unit 4503 are stored in the memory 152 or the storage device 153 as programs that can be executed by the processor 151. Further, the image storage unit 4504 and the tug storage unit 4505 are a part of the memory 152 or the storage device 153. That is, when performing the image display method according to the present example embodiment, the processor 151 functions as the calculation target extraction unit 4501, the similarity degree calculation unit 4502, the past image acquisition unit 4503, and the memory 152 or the storage device 153 functions as the image storage unit 4504 and the tug storage unit 4505. At least a part of these functions may be implemented as an electric circuit instead of a program.

[0161] In response to at least one of a time and a tug being designated (or the designation is changed) by the user, the condition acquisition unit 4101 acquires a designation condition of the time and the tug. The designation condition may be a text representing the time and the tug itself or may be a code value associated with the time and the tug.

[0162] The information acquisition unit 4102 acquires the current position and orientation of the mobile terminal 110. The definition and the acquisition method of a position and an orientation are the same as those in the first example embodiment. The information acquisition unit 4102 then uses the mobile communication device 118 to transmit information including the acquired designated condition and the current position and orientation to the server 150.

[0163] The image storage unit 4504 pre-stores past images and the positions and orientations of the past images. The image storage unit 4504 stores past images as image files that are electronic data. Each past image is provided with a unique identifier (for example, a file name). The image storage unit 4504 further stores the positions and the orientations of the past images and capturing time of past images as a table of a database, for example, in association with the identifiers of the past images.

[0164] The tug storage unit 4505 stores tugs for respective past images stored in the image storage unit 4504. The tug storage unit 4505 stores tugs for past images in association with the identifiers of the past images as a table of a database, for example. A tug is a text briefly describing the content of a past image, which may be “summer”, “festival”, “autumn leaves”, or the like as illustrated in FIG. 15, for example.

[0165] Storage of a tug for a past image may be performed at the same time as storage of the past

image in the image storage unit **4504** or may be successively performed after storage of the past image. A tug for a past image may be manually input by the user or may be automatically determined. For automatic determination of a tug, it is possible to extract an object in a past image by using the object recognition technology of the third example embodiment, for example, and automatically determine a tug related to the extracted object as a tug for the past image. When a maple tree whose leaves have turned red is included in a past image, for example, a text “autumn leaves” can be determined as a tug for the past image. Further, a plurality of tugs may be associated with a single past image.

[0166] The calculation target extraction unit **4501** extracts a past image whose similarity degree S is to be calculated from the image storage unit **4505** based on a tug of the designated condition received from the mobile terminal **110**. Specifically, first, the calculation target extraction unit **4501** acquires an identifier of a past image associated with a tug of the designated condition from the tug storage unit **4505**. The calculation target extraction unit **4501** then extracts a past image (and the position and the orientation thereof) associated with the identifier from the image storage unit **4504** as a calculation target for the similarity degree S .

[0167] The similarity degree calculation unit **4502** uses the current position and orientation received from the mobile terminal **110** and the position and orientation of the past image extracted by the calculation target extraction unit **4501** to calculate the similarity degree S indicating resemblance therebetween based on predetermined calculation criteria. The specific calculation method of the similarity degree S is the same as that in the first example embodiment.

[0168] The past image acquisition unit **4503** selects and acquires any of the past images from the image storage unit **4504** based on the similarity degree S calculated by the similarity degree calculation unit **4502** and the time of the designated condition received from the mobile terminal **110**. For example, the past image acquisition unit **4503** selects a past image closest to the time of the designated time out of past images having the similarity degree S below a predetermined threshold. Alternatively, the past image acquisition unit **4503** may use a numeric value of a difference between the time of the past image and the time of the designated condition to weight the similarity degree S and select the past image having the smallest weighted similarity degree S . The past image acquisition unit **4503** then uses the communication interface to transmit the selected past image to the mobile terminal **110**.

[0169] The image display unit **4103** displays the past image received from the server **150** on the display **113** as illustrated in FIG. 15.

[0170] The image display method according to the present example embodiment can be used in combination with the first example embodiment. In this case, a past image acquired according to a condition designated by the user may be used to start display of the difference between a current image and a past image by the image display method according to the first example embodiment. Alternatively, the image display method according to the present example embodiment may be performed alone.

[0171] While both of a time and a tug are used as an acquisition condition of a past image in the image display method according to the present example embodiment, at least one of a time and a tug may be used as an acquisition condition of a past image.

[0172] According to the present example embodiment, since a past image is acquired according to a tug or a time designated by the user, this enables the user to designate and view a past image that matches desired content or time.

[0173] FIG. 17 is a diagram illustrating a flowchart of an image display method according to the present example embodiment. The image display method is performed in cooperation with the mobile terminal **110** and the server **150**. The image display method is started when a predetermined operation such as pressing of a button is detected on the mobile terminal **110**, for example.

[0174] First, the condition acquisition unit **4101** acquires a designation condition of a time and a tug designated by the user (step **S41**). Next, the information acquisition unit **4102** acquires the

current position of the mobile terminal **110** by using the GPS receiver **117** and acquires the current orientation of the mobile terminal **110** by using the electronic compass **116** (step **S42**). The information acquisition unit **4102** uses the mobile communication device **118** to transmit information including the acquired current position and orientation to the server **150**. The order of step **S41** and step **S42** may be opposite or may be performed in parallel.

[0175] The calculation target extraction unit **4501** extracts a past image whose similarity degree **S** is to be calculated from the image storage unit **4504** based on the tug of the designated condition received from the mobile terminal **110** and tug information stored in the tug storage unit **4505** (step **S43**). The similarity degree calculation unit **4502** uses the current position and orientation received from the mobile terminal **110** to calculate the similarity degree **S** for each past image extracted by step **S43** based on Equation (1) described above, for example (step **S44**).

[0176] The past image acquisition unit **4503** acquires, from the image storage unit **4504**, one past image having the smallest similarity degree **S** out of past images whose similarity degrees **S** have been calculated in step **S44** (step **S45**). The past image acquisition unit **4503** uses the communication interface to transmit the acquired past image to the mobile terminal **110**.

[0177] The image display unit **4103** displays the past image received from the server **150** on the display **113** (step **S46**).

[0178] The processor **111** of the mobile terminal **110** and the processor **151** of the server **150** are subjects of each step (process) included in the image display method illustrated in FIG. **17**. That is, the processor **111** and the processor **151** read the program for executing the image display method illustrated in FIG. **17** from the memory **112**, the memory **152**, or the storage device **153**, execute the program to control respective units of the mobile terminal **110** and the server **150**, and thereby perform the image display method illustrated in FIG. **17**.

Other Example Embodiments

[0179] FIG. **18** is a schematic configuration diagram of the image display system **100** according to respective example embodiments described above. FIG. **18** illustrates a configuration example for the image display system **100** to implement the function of displaying an image stored in the past based on the position and orientation of a mobile terminal. The image display system **100** has the current image acquisition unit **1101** as a first image acquisition unit that acquires a first image captured by a mobile terminal; the information acquisition unit **1102** that acquires information including a position and an orientation of the mobile terminal that has captured the first image; the past image acquisition unit **1502** as a second image acquisition unit that acquires a second image based on the position and the orientation of the mobile terminal and a position and an orientation associated with a second image stored in the past in a storage device; and the difference detection unit **1103** that detects a difference between the first image and the second image.

[0180] The present invention is not limited to the example embodiments described above and can be properly changed within a scope not departing from the spirit of the present invention.

[0181] Further, the scope of each of the example embodiments includes a processing method that stores, in a storage medium, a program causing the configuration of each of the example embodiments to operate so as to realize the function of each of the example embodiments described above (more specifically, a program causing a computer to perform the method illustrated in FIG. **7**, FIG. **10**, FIG. **14**, or FIG. **17**), reads the program stored in the storage medium as a code, and executes the program in a computer. That is, the scope of each of the example embodiments also includes a computer readable storage medium. Further, each of the example embodiments includes not only the storage medium in which the program described above is stored but also the program itself.

[0182] As the storage medium, for example, a floppy (registered trademark) disk, a hard disk, an optical disk, a magneto-optical disk, a CD-ROM, a magnetic tape, a nonvolatile memory card, or a ROM can be used. Further, the scope of each of the example embodiments includes an example that operates on OS to perform a process in cooperation with another software or a function of an

add-in board without being limited to an example that performs a process by an individual program stored in the storage medium.

[0183] The whole or part of the example embodiments disclosed above can be described as, but not limited to, the following supplementary notes.

Supplementary Note 1

[0184] An image display system comprising: [0185] a first image acquisition unit configured to acquire a first image captured by a mobile terminal; [0186] an information acquisition unit configured to acquire information including a position and an orientation of the mobile terminal that has captured the first image; [0187] a second image acquisition unit configured to, based on the position and the orientation of the mobile terminal and a position and an orientation associated with a second image stored in the past in a storage device, acquire the second image; and [0188] a difference detection unit configured to detect a difference between the first image and the second image.

Supplementary Note 2

[0189] The image display system according to supplementary note 1 further comprising an image display unit configured to display the difference in addition to at least one of the first image and the second image.

Supplementary Note 3

[0190] The image display system according to supplementary note 2, wherein the image display unit displays a frame surrounding a region including the difference on the first image and the second image.

Supplementary Note 4

[0191] The image display system according to any one of supplementary notes 1 to 3, wherein the position and the orientation associated with the second image are a position and an orientation of a captured range of the second image.

Supplementary Note 5

[0192] The image display system according to any one of supplementary notes 1 to 4 further comprising a similarity degree calculation unit configured to calculate a similarity degree indicating resemblance of the position and the orientation of the mobile terminal to the position and the orientation associated with the second image, [0193] wherein the second image acquisition unit acquires the second image based on the similarity degree.

Supplementary Note 6

[0194] The image display system according to any one of supplementary notes 1 to 5 further comprising: [0195] the mobile terminal having the first image acquisition unit and the information acquisition unit; and [0196] a server having the second image acquisition unit, [0197] wherein the difference detection unit is provided either in the mobile terminal or the server.

Supplementary Note 7

[0198] The image display system according to any one of supplementary notes 1 to 6 further comprising a peripheral information display unit configured to display the position associated with the second image present within a predetermined distance from the position of the mobile terminal on a map.

Supplementary Note 8

[0199] The image display system according to any one of supplementary notes 1 to 7 further comprising a proximity notification unit configured to perform notification when the second image is present within a predetermined distance from the position of the mobile terminal.

Supplementary Note 9

[0200] The image display system according to any one of supplementary notes 1 to 8 further comprising an object recognition unit configured to recognize an object included in the first image and the second image and output information on the object.

Supplementary Note 10

[0201] The image display system according to any one of supplementary notes 1 to 9 further comprising a condition acquisition unit configured to acquire a condition designated by a user of the mobile terminal, [0202] wherein the second image acquisition unit acquires the second image based on the designated condition in addition to the position and the orientation of the mobile terminal and the position and the orientation associated with the second image.

Supplementary Note 11

[0203] A terminal comprising: [0204] a first image acquisition unit configured to acquire a first image captured; [0205] an information acquisition unit configured to acquire information including a position and an orientation; and [0206] a difference detection unit configured to receive a second image acquired based on the position and the orientation and a position and an orientation associated with the second image stored in the past in a storage device and detect a difference between the first image and the second image.

Supplementary Note 12

[0207] A method comprising: [0208] acquiring a first image captured by a mobile terminal; [0209] acquiring information including a position and an orientation of the mobile terminal that has captured the first image; [0210] based on the position and the orientation of the mobile terminal and a position and an orientation associated with a second image stored in the past in a storage device, acquiring the second image; and [0211] detecting a difference between the first image and the second image.

Supplementary Note 13

[0212] A program configured to cause a computer to execute: [0213] acquiring a first image captured by a mobile terminal; [0214] acquiring information including a position and an orientation of the mobile terminal that has captured the first image; [0215] based on the position and the orientation of the mobile terminal and a position and an orientation associated with a second image stored in the past in a storage device, acquiring the second image; and [0216] detecting a difference between the first image and the second image.

Claims

1. An information processing apparatus comprising: at least one memory configured to store computer program code; and at least one processor configured to execute the computer program code to: acquire a first image that is captured by a camera mounted on a mobile object, and a second image (past image) that was captured by the camera mounted on the mobile object; and acquire a position of the camera and a capturing direction of the camera, wherein in a case where the mobile object acquiring the first image is moved to a position at which the second image was acquired, positioning of the first image and the second image is performed based on features of the first image and the second image.
2. The information processing apparatus according to claim 1, wherein the at least one processor is further configured to execute the computer program code to accept designation of a region as a target of object recognition on the second image.
3. The information processing apparatus according to claim 1, wherein the at least one processor is further configured to execute the computer program code to perform notification to a user in a case where a current position of the camera and the position of the camera that captured the second image are within a predetermined distance.
4. The information processing apparatus according to claim 1, wherein the at least one processor is further configured to execute the computer program code to display a difference between the first image and the second image in a case where the camera is moved within a predetermined distance from the position of the camera that captured the second image.
5. The information processing apparatus according to claim 4, wherein the at least one processor is further configured to execute the computer program code to display a frame surrounding a region

including the difference on the first image or the second image.

6. The information processing apparatus according to claim 1, wherein the at least one processor is further configured to execute the computer program code to calculate a similarity degree between the first image and the second image.

7. An information processing method comprising: acquiring a first image that is captured by a camera mounted on a mobile object, and a second image (past image) that was captured by the camera mounted on the mobile object; and acquiring a position of the camera and a capturing direction of the camera, wherein in a case where the mobile object acquiring the first image is moved to a position at which the second image was acquired, positioning of the first image and the second image is performed based on features of the first image and the second image.

8. The information processing method according to claim 7, further comprising: accepting designation of a region as a target of object recognition on the second image.

9. The information processing method according to claim 7, further comprising: performing notification to a user in a case where a current position of the camera and the position of the camera that captured the second image are within a predetermined distance.

10. The information processing method according to claim 7, further comprising: displaying a difference between the first image and the second image in a case where the camera is moved within a predetermined distance from the position of the camera that captured the second image.

11. The information processing method according to claim 10, further comprising: displaying a frame surrounding a region including the difference on the first image or the second image.

12. The information processing method according to claim 7, further comprising: calculating a similarity degree between the first image and the second image.

13. A non-transitory computer-readable storage medium storing a program that causes a computer to perform: acquiring a first image that is captured by a camera mounted on a mobile object, and a second image (past image) that was captured by the camera mounted on the mobile object; and acquiring a position of the camera and a capturing direction of the camera, wherein in a case where the mobile object acquiring the first image is moved to a position at which the second image was acquired, positioning of the first image and the second image is performed based on features of the first image and the second image.

14. The non-transitory computer-readable storage medium according to claim 13, further storing a program that causes the computer to perform: accepting designation of a region as a target of object recognition on the second image.

15. The non-transitory computer-readable storage medium according to claim 13, further storing a program that causes the computer to perform: performing notification to a user in a case where a current position of the camera and the position of the camera that captured the second image are within a predetermined distance.

16. The non-transitory computer-readable storage medium according to claim 13, further storing a program that causes the computer to perform: displaying a difference between the first image and the second image in a case where the camera is moved within a predetermined distance from the position of the camera that captured the second image.

17. The non-transitory computer-readable storage medium according to claim 16, further storing a program that causes the computer to perform: displaying a frame surrounding a region including the difference on the first image or the second image.

18. The non-transitory computer-readable storage medium according to claim 13, further storing a program that causes the computer to perform: calculating a similarity degree between the first image and the second image.
